Chapter 3 — Why $\sqrt{d_k}$?

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Type: theory + notebook
Ordering: theory-first
Depends on: 1 (scaled dot-product attention), 2 (softmax & temperature)
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3.1 Motivation

The scaling factor $1/\sqrt{d_k}$ in scaled dot-product attention is not cosmetic. It is the factor that keeps pre-softmax scores of order one as the head dimension grows. Without it, softmax saturates at initialization and gradients die. Why $\sqrt{d_k}$ specifically — not d_k, not $\log d_k$, not a learned scalar — is a variance argument followed by a gradient argument.

3.2 Storyline

The mechanism is a three-link chain.

1. **Variance in, variance out.** The inner product of two independent random vectors with i.i.d. unit-variance entries has variance equal to the dimension. Raw scores $s_{ij} = \langle q_i, k_j \rangle$ therefore have standard deviation $\sqrt{d_k}$, which grows unboundedly with head dimension. For $d_k = 128$, scores live on the scale of ± 11.
2. **Softmax concentrates under large inputs.** From Chapter 2, softmax with a winner of margin Δ gives $p_{\text{win}} = e^{\Delta}/(e^{\Delta} + (n-1))$; once $\Delta \gg \log(n-1)$, $p_{\text{win}} \to 1$ and the distribution is effectively one-hot. Large-scale inputs push softmax into this saturated regime.
3. **Saturated softmax has zero Jacobian.** $\partial p_i / \partial s_j = p_i(\delta_{ij} - p_j)$. When p is near one-hot every entry of this Jacobian is near zero. No gradient flows back to Q, K, so attention cannot learn.

Dividing by $\sqrt{d_k}$ rescales the first link so pre-softmax inputs have unit variance regardless of head dimension. Softmax then sees inputs of moderate magnitude and the gradient survives.

Regime notes. The argument is sharpest at initialization. After training, learned W_Q, W_K restructure the score distribution in ways the i.i.d. hypothesis does not capture (see §3.3, Empirical observation 3.4). Measured pre-softmax score variance in a trained model is typically below one even with the $\sqrt{d_k}$ factor present; the factor's job is to keep early training healthy, not to maintain unit variance throughout. The factor is a fixed constant — no train / val / inference distinction, no interaction with dropout or masking.

3.3 Rigorous core

Lemma 3.1 (Variance of an inner product). Let $q, k \in \mathbb{R}^{d_k}$ have independent components q_i, k_i with $\mathbb{E}[q_i] = \mathbb{E}[k_i] = 0$ and $\text{Var}(q_i) = \text{Var}(k_i) = 1$, and $q \perp k$ as random vectors. Then

$$\mathbb{E}[\langle q, k \rangle] = 0, \qquad \text{Var}(\langle q, k \rangle) = d_k.$$

Sketch: mean by independence; variance by independence across i and the factoring $\mathbb{E}[q_i^2 k_i^2] = \mathbb{E}[q_i^2]\mathbb{E}[k_i^2] = 1$. Generalized in Exercise 3.1.

Proposition 3.2 (The $\sqrt{d_k}$ correction). Under the hypotheses of Lemma 3.1, $\text{Var}(\langle q, k \rangle / \sqrt{d_k}) = 1$.

Lemma 3.3 (Jacobian vanishing under saturation). Let $p = \text{softmax}(s)$ for $s \in \mathbb{R}^n$. Let $J_{ij} = \partial p_i / \partial s_j = p_i(\delta_{ij} - p_j)$. If $p \to e_k$ (one-hot at index k), then $\|J\|_F \to 0$. Sketch: case-split on $i = k$ vs. $i \neq k$ — every entry vanishes individually.

Corollary 3.3.1. Pre-softmax scores with standard deviation $\sqrt{d_k}$ place softmax in the saturating regime (Chapter 2, §2.4); combined with Lemma 3.3, this means gradients at W_Q, W_K through an unscaled attention layer vanish in expectation as $d_k \to \infty$ at initialization.

Empirical observation 3.4 (The i.i.d. hypothesis breaks post-training). In a trained model the entries of q, k are not i.i.d. — learned W_Q, W_K concentrate mass in specific directions dictated by the task. Pre-softmax score variance in trained attention is typically below one even with the $\sqrt{d_k}$ factor present. The scaling argument is sharp at init; after training it is conservative. This motivates alternatives like QK-norm (Exercise 3.3) that do not depend on the i.i.d. hypothesis.

3.4 Numerical example — variance scaling is a distributional fact

A single hand computation of $\langle q, k \rangle$ for fixed q, k doesn't surface what Lemma 3.1 is saying — the lemma is about the distribution of the inner product across i.i.d. samples, not any particular value. The minimal experiment that reveals the mechanism is a Monte-Carlo estimate of the variance across d_k:

```
import numpy as np

M = 100_000
for d_k in [1, 4, 16, 64, 256, 1024]:
    q = np.random.randn(M, d_k)
    k = np.random.randn(M, d_k)
    s = np.sum(q * k, axis=1)
    print(f"d_k={d_k:>5}  var(s) ≈ {s.var():8.2f}   var(s/√d_k) ≈ {(s / np.sqrt(d_k)).var():5
```

Expected: the first column tracks d_k linearly, the second stays near 1. The notebook sweeps this across several random seeds and measures the full distribution, not just the variance.

3.5 Mechanics check

- **Scale by** $1/\sqrt{d_k}$ — keeps pre-softmax score variance at $O(1)$ regardless of head dimension (Lemma 3.1 + Proposition 3.2). Without this, softmax saturates at init and gradients die.
- **Why** $\sqrt{d_k}$**, not** d_k — we want standard deviation $O(1)$, not variance $O(1/d_k)$. Dividing by d_k shrinks scores too hard and collapses softmax toward uniform (temperature $\to \infty$ in Chapter 2's sense), which is also uninformative.
- **Why not a learned scalar at init** — a learned scalar is initialized somewhere; starting far from $1/\sqrt{d_k}$ creates the saturation problem this factor is designed to prevent. Learned scalars are a post-init tool (see QK-norm, Exercise 3.3).
- **Interaction with softmax, not with masking or dropout** — the factor sits between $QK^\top$ and softmax; it does not interact with the additive mask (which is applied to the scaled scores) or with attention dropout (which is applied to the softmax output). No train / val / inference asymmetry.

3.6 Exercises

3.1 ⋆⋆ [Mechanism dissection — general variances.] Let entries of q, k be i.i.d. with $\mathbb{E}[q_i] = \mathbb{E}[k_i] = 0$, $\mathrm{Var}(q_i) = \sigma_q^2$, $\mathrm{Var}(k_i) = \sigma_k^2$. Derive the scaling factor $c(d_k, \sigma_q, \sigma_k)$ such that $\langle q, k \rangle / c$ has unit variance. What does this say about the common convention of initializing both W_Q and W_K at the same scale?

3.2 ⋆⋆ [Failure-mode construction — dependent q and k.] Suppose $q = W_Q x$ and $k = W_K x$ for the same input $x \in \mathbb{R}^d$, so q and k are not independent. Let x have i.i.d. unit-variance entries. Derive $\mathrm{Var}(\langle q, k \rangle)$ in terms of W_Q, W_K. Does it still scale like d_k? Construct explicit W_Q, W_K for which the variance is $\Theta(d_k^2)$ (much larger than d_k), and others for which it is $O(1)$ (much smaller). What structural feature of $W_Q^\top W_K$ controls the answer?

3.3 ⋆⋆⋆ [Theorem about the numerics — QK-norm.] QK-norm replaces $QK^\top / \sqrt{d_k}$ with $\gamma \cdot \hat{Q}\hat{K}^\top$, where $\hat{q}_i = q_i / \|q_i\|_2$ row-wise (similarly $\hat{k}_j$), and $\gamma \in \mathbb{R}$ is learned.

(a) Prove that $\hat{q}^\top \hat{k} \in [-1, 1]$ deterministically, independent of any distributional assumption on q, k.
(b) Conclude that pre-softmax scores are bounded by $\pm\gamma$, and contrast this with the $\sqrt{d_k}$ guarantee (which is an expectation, not a bound).
(c) In Chapter 2's language, what quantity does γ play the role of? State the analogy precisely.
(d) Give one situation in which QK-norm is strictly more robust than $\sqrt{d_k}$-scaling, and one in which it is strictly less expressive.

3.4 ⋆⋆⋆ [Predict-then-verify — unscaled attention across d_k.] The notebook will train a 1-layer transformer with unscaled attention on a simple copy task, sweeping $d_k \in \{4, 16, 64, 256\}$. For each d_k, predict before running which of the following occurs, and justify from Lemmas 3.1 and 3.3 and Corollary 3.3.1:

(a) training loss fails to decrease at all (plateau at the init-entropy floor);
(b) training loss decreases slowly and converges to a worse optimum than the scaled baseline;
(c) training loss decreases normally (the unscaled model learns fine);
(d) training loss diverges (blows up) after some number of steps.

Write your prediction as a 4×4 table (rows: d_k, columns: outcomes (a)–(d), entries: your assigned probability summing to 1 per row) with a one-line reason per row. State your overall confidence (low / medium / high, with reason).

3.7 Before the notebook (`03_variance_scaling.ipynb`)

Write down the following predictions before opening the notebook. The notebook's purpose is to test them.

1. **Empirical variance of unscaled scores** as a function of $d_k \in \{1, 4, 16, 64, 256, 1024\}$ — shape, scaling exponent, numeric values at $d_k = 1$ and $d_k = 1024$.
2. **Empirical variance of scaled scores** — shape, scaling exponent. State the expected deviation from 1 (concentration rate in M, the number of Monte-Carlo samples).
3. **Softmax entropy** of a single row as a function of d_k, for unscaled vs. scaled inputs, with $n = 32$ keys. State the limit of each as $d_k \to \infty$ in one word.
4. **Max / mean softmax ratio** in a single row, as a function of d_k, for unscaled vs. scaled. Direction and limit.

5. **Frobenius norm of the softmax Jacobian** (evaluated at a random score vector) as a function of d_k, unscaled vs. scaled. Direction and the approximate rate of decay in the unscaled case (polynomial or exponential in d_k?).

For each prediction: direction, order-of-magnitude estimate, confidence level with a one-line reason. The notebook confirms or refutes each in its Finding section.